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Advanced Insurance Risk Modeling for New Customers Using Balanced Ensembles and Transformer Architectures

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Abstract: Classifying new customers in insurance portfolios is particularly challenging due to the absence of historical behaviour, extreme class imbalance, heavy-tailed loss distributions, and strict operational constraints. Traditional machine learning methods, including the baseline approach proposed in previous studies, often struggle to detect high-risk clients because they optimize general predictive accuracy rather than business-critical outcomes. In this work, we extend the existing methodology by evaluating two complementary business-aware classification strategies: (i) a balanced bagging ensemble specifically designed to address class imbalance and optimize expected profit under customer-omission constraints, and (ii) a Transformer-based architecture capable of capturing richer representations of customer features. Both models incorporate the asymmetric financial cost structure of insurance risk and explicitly respect operational selection limits. Using a proprietary large-scale auto insurance dataset comprising 51,618 customers, and validated on nine synthetic datasets, we compare these three approaches using both statistical tests (ANOVA, Friedman, pairwise comparisons) and business metrics. Our results show that both new approaches substantially outperform the baseline methodology ($p < 0.001$), with the balanced ensemble achieving the most favourable trade-off between predictive performance, robustness, and computational efficiency, while the Transformer provides competitive performance with enhanced generalization on perturbed data. This study contributes to business-optimized machine learning and demonstrates two practical, scalable solutions for risk-driven decision-making in real-world insurance environments.

Keywords: insurance risk modeling; new customer classification; high-risk customer detection; balanced ensembles; transformer models

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1. Introduction

Accurate customer risk assessment is essential in the insurance industry to maintain profitability and competitiveness. Misclassification of customers can cause considerable financial losses, as poor risk assessment often leads to underpriced premiums and rising claims costs [1]. Predicting the performance of new customers is particularly difficult, as historical data is scarce and the costs of misclassification are high [2]. Customer classification is further complicated by severe class imbalance. Loss-generating customers typically account for only 10% to 20% of the portfolio but are responsible for 60% to 80% of total losses [3]. In this study, the problem is explicitly defined as a binary classification task under business constraints, where new customers are assigned to either a high-risk or a low-risk class. While the model produces probability scores, these are only used for thresholding under business constraints. Still, our final objective is categorical classification rather than continuous risk estimation. Conventional approaches tend to favor the majority class, resulting in poor identification of high-loss customers [4]. Business constraints add

to the challenge, since insurers cannot omit large numbers of customers without risking business volume and market share [5].

Most existing work addresses these issues through feature selection and model optimization. Soriano-Gonzalez et al. [6], for example, predicted new customer performance using XGBoost and LightGBM, combined with extensive feature engineering and profitability thresholds. However, their method does not directly address class imbalance or asymmetric misclassification costs [7]. It relies on arbitrary economic thresholds and does not incorporate operational constraints such as acceptable rejection rates or business risk tolerance. Nor does it explore ensemble strategies tailored to imbalance or apply business-driven principles such as the Pareto rule for threshold selection [8].

To overcome these limitations, this study evaluates two complementary business-aware methodologies: (i) a balanced bagging ensemble to enforce class balance and optimize profit under customer-omission constraints, and (ii) a Transformer-based classifier capable of learning richer feature representations and benefiting from augmented training data. The balanced ensemble approach preserves all high-risk customers while systematically undersampling the majority class according to a profit-optimized sampling ratio. Multiple XGBoost base learners are trained on these balanced subsets, and their outputs are aggregated to improve robustness, align predictions with asymmetric costs, and comply with the maximum 8% customer-omission constraint. In contrast, the Transformer model adopts a strategy based on deep representation learning. During training, the model increases data diversity through a predefined augmentation ratio and selects, at each epoch, the checkpoint that maximizes validation profit. Final decisions for new customers are obtained by applying probability thresholds that comply with the operational constraints of the business. These two approaches—one focused on structured imbalance handling and the other on deep representation learning—offer distinct yet complementary solutions to the challenge of assessing the risk of new customers in insurance portfolios.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data preparation process, including preprocessing, feature selection, and statistical analysis. Section 4 outlines the proposed methodology, incorporating business constraints and the balanced bagging ensemble approach. Section 5 presents the experimental results and discussion, including model performance, economic impact, and comparisons with baseline methods. Finally, Section 7 summarizes the conclusions and outlines directions for future work.

2. Related Work

Machine learning methods are increasingly used in the insurance sector to support risk assessment, underwriting, and claims management. Their ability to process large datasets and support tasks such as classification and prediction has been well documented in various financial applications [9–11]. In life and auto insurance, models such as logistic regression, decision trees, random forest, support vector machines, naive Bayes, and XGBoost have been used to predict customer risk levels and claim outcomes [12,13]. These models reduce manual work in underwriting and improve operational efficiency. In the context of auto insurance, studies have shown that tree-based ensemble methods outperform traditional general linear models, particularly for predicting claim frequency and severity in imbalanced datasets [14]. Ensemble methods such as bagging, boosting, and stacking have proven effective in improving classification accuracy across various insurance applications, including fraud detection and high-cost claim prediction [15–17]. Techniques such as XGBoost and explainable boosting machines have been applied to maintain both high accuracy and model transparency [18]. For instance, Soriano-Gonzalez et al. [6] proposed a profit-based objective to address asymmetric misclassification costs in insurance classification. However, their approach had several limitations. They did not apply techniques to address class imbalance, lacked control over customer omission rates, and used fixed thresholds without systematic tuning. Their model was also limited to a single boosting algorithm, reducing its flexibility and robustness.

Several studies have focused on imbalanced data problems, as events like fraud or severe claims are rare compared to the broader customer base [19]. To address this, researchers have applied undersampling, oversampling (e.g., SMOTE [20]) and more recent GAN-based oversampling approaches [21], as well as class weighting and adaptive ensemble methods. Beyond these general strategies, ensemble-based imbalance methods have also been proposed, including SMOTEBoost [20], RUSBoost [22], Balanced Random Forest [23], and EasyEnsemble2 [24]. These approaches combine boosting or bagging with resampling to improve minority detection and have been widely adopted in imbalanced learning tasks. However, they do not incorporate explicit business constraints or profit-driven objectives, which are central to our framework.

Baran and Rola [25] and Khamesian et al. [26] combined decision trees or neural networks with resampling techniques to improve the detection of rare insurance events. Similar challenges are present in medical applications, where rare event prediction has high ethical and operational implications [27]. Nonetheless, many studies optimize mainly for statistical measures such as AUC, which reflects ranking performance rather than direct classification quality, or F1-score, which is more appropriate under class imbalance. However, these metrics often ignore business constraints such as asymmetric costs or acceptance limits. For example, thresholds are often set at $p = 0.5$ with no justification based on cost, regulation, or operational targets. Some approaches from other domains, such as online recommendation systems, have applied reinforcement learning and multitask neural networks to optimize conversion rates [28], but these are mostly limited to digital channels. In contrast, profit-based objectives in insurance remain underexplored, although some work has looked into economic performance in lapse risk modeling [29]. Our study builds on this gap by introducing a balanced bagging ensemble that includes multiple classifiers to manage severe class imbalance while preserving the distribution of rare classes. We integrate an explicit constraint to cap customer omission at 8%, ensuring operational feasibility. The sampling ratio (r) is optimized using stratified cross-validation and sensitivity analysis, and majority voting is applied to improve model robustness and generalization.

Beyond risk prediction, ML has also supported tasks such as customer segmentation. Clustering techniques including k-means, hierarchical clustering, and DBSCAN have been used to group customers with similar characteristics, supporting targeted marketing and service strategies [30]. Nevertheless, classification of new customers with scarce historical information remains limited, especially when data acquisition is costly or incomplete [31]. In such settings, the financial impact of misclassification can be high, but few studies provide solutions to handle this effectively [32]. Recent developments in deep learning have further expanded machine learning applications in insurance. Convolutional and recurrent neural networks, including long short-term memory variants, have been used for processing unstructured data, such as images of damaged vehicles or sequential customer records [33–35]. Natural language processing has also been integrated with machine learning to analyze text from policy documents, claims, and customer interactions [36–38]. Despite these advances, model interpretability remains a challenge, especially for deep learning. Insurers need models that are not only accurate but also explainable to meet regulatory requirements and maintain user trust [39,40]. Tools such as SHAP (Shapley additive explanations) and LIME have been adopted to improve the transparency of model decisions [41,42]. Still, SHAP values provide estimates of feature importance and their reliability can decrease in high-dimensional settings, where heuristics are often used in practice instead of exact attribution. In parallel, feature selection strategies tailored to insurance, such as ordinal risk categories, have also been explored [43]. Another ongoing concern is data privacy and compliance with regulations such as the General Data Protection Regulation in the European Union [44,45]. These constraints must be considered in the practical deployment of machine learning models in insurance workflows.

Recent advances have applied deep learning and feature-interaction models to tabular insurance data. Transformer-based models (e.g., TabTransformer, FT-Transformer)

capture complex dependencies across features [46], and TabNet has shown competitive results in public benchmarks [47]. Nonetheless, these methods typically require substantial computational resources and offer limited interpretability, which is a key requirement in regulated insurance contexts. In practice, post-hoc attribution can be computationally intensive and is often applied outside the training loop. Moreover, while deep learning models can achieve strong predictive performance, their application in insurance remains constrained by transparency requirements, regulatory obligations, and the operational need for models that can be audited and explained. As a result, research has increasingly focused on balancing predictive accuracy with interpretability, motivating continued exploration of architectures capable of modeling complex tabular interactions while remaining suitable for real-world deployment [14,48].

To provide a structured overview, Table 1 summarizes the main state-of-the-art (SOTA) techniques most frequently applied to imbalance and tabular learning problems, highlighting their strengths, limitations in insurance contexts, and how our proposed approach differs. This comparative perspective emphasizes that, unlike existing methods, our framework directly integrates business constraints and profit-driven optimization, while maintaining interpretability and practical feasibility.

Table 1. Comparison of SOTA imbalance methods with the proposed approach.

Method	Strengths	Limitations	Difference vs. Our Approach
SMOTE / GAN oversampling	Increase minority samples via synthesis [20,21].	May create unrealistic cases; poor fit for heavy-tailed data.	Keeps only real cases, no synthetic artifacts.
Cost-sensitive / Focal Loss	Penalize costly misclassifications; improve recall [31,32].	Ignore business limits (e.g., $\leq 8\%$ omission).	Direct integration of operational constraints.
TabNet (deep learning)	Captures feature interactions; strong benchmarks [47].	High cost; low interpretability; limited regulatory use.	Interpretable tree ensembles, easy deployment.
Transformer models	Model complex, high-order dependencies [46].	Require large data; expensive; low transparency.	Balanced trade-off of accuracy and interpretability.
SHAP-based selection	Nonlinear feature importance; interpretability [41,42].	Heavy computation; post-hoc only.	Mutual information used within training pipeline.
SMOTEBoost / RUSBoost / Balanced RF / EasyEnsemble2	Combine boosting/bagging with resampling; widely used [20,22,23,49].	Do not preserve real minority; lack profit/business focus.	Preserves genuine minority, profit-optimized, business limits enforced.
Proposed ensemble	Preserves minority; profit-driven; interpretable.	Slightly lower precision; higher cost than single models.	Directly aligned with insurance business and regulation.

We adopt a balanced bagging ensemble with profit-driven optimization because, unlike synthetic oversampling or purely cost-sensitive methods, this approach preserves the authenticity of minority cases, directly incorporates business constraints ($x \leq 8\%$ omission), and provides interpretable results suitable for regulated insurance contexts. Additionally, we include a Transformer-based model to complement the ensemble approach, allowing us to benchmark a modern deep-learning architecture within the same business-constrained framework. Rather than replacing traditional methods, the Transformer serves as a high-capacity alternative whose performance can be contrasted against an interpretable ensemble solution. This enables a broader evaluation of methodological trade-offs, balancing interpretability and representational power when addressing new-customer risk classification in insurance.

3. Data Preparation

The empirical analysis uses a real-world insurance dataset with 116,934 customer records covering 2016 to 2023. The raw dataset initially contained 247 features on demographics, vehicle attributes, policy details, and historical claims. To ensure applicability to prospective customers, we removed variables not available at acquisition (e.g., post-policy information and claim outcomes).

3.1. Statistical Analysis

We tested the distributional properties of the continuous target variable, defined as the customer's average annual benefit (profits minus claims, AAP). The null hypothesis assumed that the customer's annual benefit follows a standard probability distribution (normal, exponential, logistic, or Gumbel). The Jarque–Bera test [50] (1.81×10^8, $p < 0.001$) rejects normality, and Anderson–Darling tests [51] reject normal, exponential, logistic, and Gumbel probability distributions at all significance levels. Losses show heavy-tailed behavior, concentrating business risk. Of $51,618$ customers, $9,984$ (19.3%) generated negative returns, with total losses of $11,328,603$ euros, while the portfolio as a whole produced only $909,881$ euros in net benefit before intervention. These figures result from aggregating individual returns: the cumulative claims of loss-generating customers exceeded their premiums by 11.3 million euros, whereas across the full portfolio premiums only outweighed claims by 0.9 million euros. Thus, loss-generating customers impose losses more than 12 times the net benefit, underscoring the need for accurate identification.

Following Clauset et al. [52], we conducted a power-law analysis of the tail behavior of customer annual losses, defined as negative values of the annual benefit distribution. Maximum likelihood estimation gave an exponent $\alpha = 2.381 \pm 0.057$ with $x_{\min} = 3,454$ euros. The Kolmogorov–Smirnov test ($D = 0.0197$, with $p = 0.652$) indicated good fit. Similarly, semi-parametric bootstrap tests did not reject the power-law hypothesis. Likelihood ratio tests favored the power law over the exponential ($R = 191.085$, with $p < 0.001$) and found no significant difference from the lognormal probability distribution ($R = -0.040$, with $p = 0.859$). The exponent lies in the critical regime $2 < \alpha < 3$, implying a finite mean but divergent theoretical variance, reflected in extreme sensitivity to tail values and unstable variance estimates. These features undermine conventional machine learning, leading to: (i) violations in methods assuming stable variance; (ii) under-representation of high-impact tail events in random samples; and (iii) unreliable extrapolation by synthetic oversampling methods. The proposed approaches, both the ensemble and the Transformer, address these limitations by maintaining a consistent representation of extreme cases in the data.

Table 2. Loss concentration across multiple thresholds.

Cumulative Loss	Customer Percentage	95% CI
70%	13.5%	(11.8%, 15.1%)
75%	17.4%	(15.6%, 19.1%)
80%	22.5%	(20.7%, 24.4%)
85%	29.4%	(27.5%, 31.2%)
90%	38.5%	(36.7%, 40.2%)

Moreover, bootstrap analysis with replacement using $1,000$ iterations confirmed extreme loss concentration: 22.5% of loss-generating customers account for 80% of total losses, with a 95% confidence interval (CI) of (20.7%, 24.4%). The bootstrap procedure used simple random sampling with replacement from the original dataset, with each bootstrap sample maintaining the original dataset size (n = $9,984$). As shown in Table 2 and Figure 1, a minority of customers drive most losses, reinforcing the need to prioritize minority-class identification over global classification accuracy.

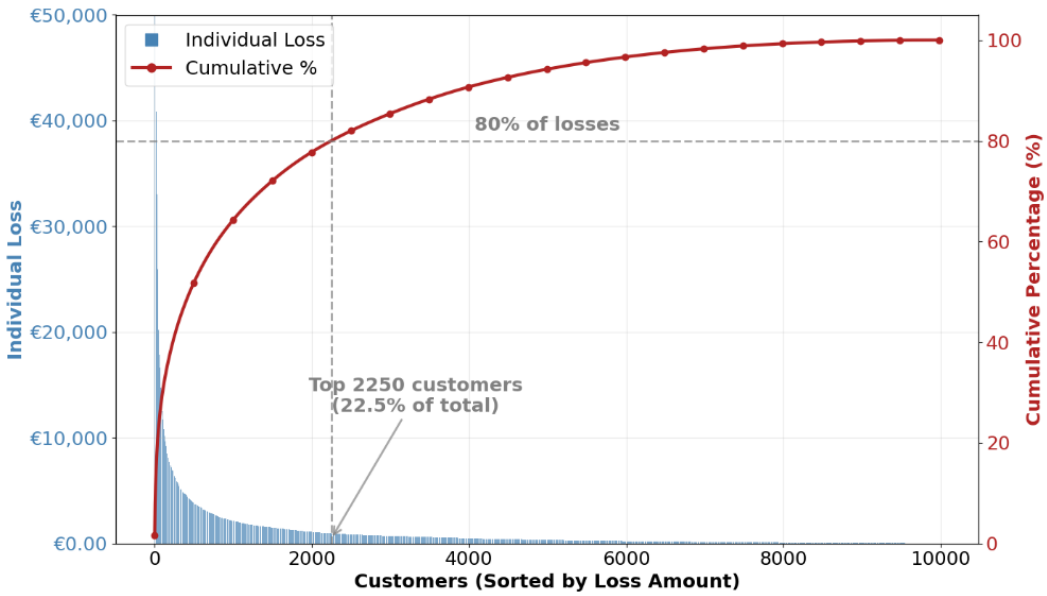

Figure 1. Cumulative loss distribution. Blue bars show individual losses, red curve the cumulative percentage. The top 22.5% of customers account for 80% of total losses.

3.2. Preprocessing & Feature Selection

Following Soriano-Gonzalez et al. [6], we applied a standard preprocessing pipeline that included one-hot encoding of categorical variables and the standardization of column names to ensure computational compatibility. Missing values were imputed using SoftImputer, implemented in the `fancyimpute` Python package, which applies iterative matrix completion with nuclear norm regularization. Customers lacking data from SINCO (the sector-wide motor insurance history file managed by TIREA [53], which provides access to Spain's Historical Automobile Insurance Information System, SIHSA) were omitted, leaving 51,618 observations with 196 features. This exclusion was necessary because SINCO variables provide the only standardized information on prior insurance history and claims, which are critical predictors of customer risk. Without this data, the model would face severe information loss and potential bias in risk classification, so we retained only customers with complete SINCO records.

For the ensemble method, feature selection was performed using mutual information instead of correlation analysis. Mutual information scores were computed for all 194 inputs with respect to the binary classification target, and the 100 features with the highest scores were retained based on their predictive relevance for identifying high-risk customers. This process removed 53 features with very low predictive information (MI scores < 0.001). In contrast to correlation-based approaches, mutual information accounts for non-linear relationships and links feature selection directly to the classification goal while maintaining interpretability. However, for the Transformer model, applying feature selection resulted in a notable drop in performance, so all 194 input features were used.

The dataset was partitioned into three independent subsets using stratified sampling to preserve representative class distributions: training (56%), validation (14%), and test (30%). For the ensemble models, the training set was used to fit models under different sampling ratio configurations, and the validation set was used to get the optimal sampling ratio r via profit-based evaluation. For the Transformer model, the training set was used for parameter fitting, while the validation set served to monitor and control overfitting during training, and to select the best-performing model in the training phase based on validation profit. The test set was kept completely separate and used only for the final unbiased assessment of the selected configuration in both approaches.

4. Methodology

4.1. Balanced Ensemble Model

4.1.1. Binary Classification under Business Constraints

The binary classification operates under strict business constraints reflecting real-world deployment conditions. The insurance company can omit at most 8% of potential customers due to competitive positioning, regulatory requirements, and acquisition targets. Rather than tuning this threshold to maximize validation metrics, which risks overfitting, we adopt the business-imposed constraint directly. The classification threshold is set at the $8th$ percentile of the target variable (customer's average annual benefit), corresponding to values below $-1,708$ euros, emphasizing generalizability over dataset-specific performance gains.

This constraint produces severe class imbalance with direct implications for model choice. We flag $4,129$ customers (8% of the dataset) as high-loss cases, yielding a minority-to-majority ratio of $1:11.6$. Traditional classifiers achieved high overall accuracy ($85-92\%$) but failed to identify high-risk customers. Random Forest achieved 91.7% accuracy yet detected only 0.2% of high-risk customers (1 out of 595 cases). Logistic Regression showed similar performance with 91.6% accuracy but only 0.5% recall for the minority class. These results highlight that accuracy scores are misleading when they hide poor performance on critical minority classes.

The $1:11.6$ imbalance ratio creates three key problems. First, algorithms are overwhelmed by majority class patterns; a dummy classifier predicting only the majority class achieved 91.8% accuracy while never identifying a single high-risk customer. Second, even sophisticated methods struggle to balance precision and recall. Both Random Forest and Logistic Regression achieved reasonable precision (20%) but poor recall ($< 1\%$) for high-risk cases. Third, these detection failures have severe business consequences: standard classifiers captured only $8.5-13.8\%$ of maximum possible benefit, compared to 27.2% with improved minority class detection.

The costs of misclassification are highly asymmetric. Missing high-loss customers (false negatives) leads to direct financial losses, while incorrectly flagging profitable customers (false positives) creates opportunity costs but no direct losses. This cost asymmetry prioritizes minority class recall over overall accuracy, leading to our balanced ensemble approach that achieved 31% recall while staying within operational limits.

4.1.2. Balanced Bagging Ensemble Approach

Our approach builds upon ensemble methods for class imbalance learning [54], introducing novel elements for heavy-tailed financial risk assessment. While methods such as EasyEnsemble [24] and SMOTEBoost [20] have demonstrated effectiveness in general imbalanced scenarios, the specific requirements of financial risk assessment—extreme loss concentration, business constraints, and asymmetric costs—motivate our specialized approach.

The theoretical motivation for balanced bagging in heavy-tailed loss scenarios stems from three principles. First, in power-law distributed losses ($\alpha = 2.381$), accurate minority class identification yields disproportionate business value; 22.6% of loss-generating customers account for 80% of total losses. Second, unlike synthetic oversampling, our approach maintains genuine characteristics of minority class instances, crucial for heavy-tailed distributions where extreme tail behavior cannot be reliably synthesized without introducing distributional artifacts. Third, by training each base learner on different majority class samples while maintaining complete coverage of the minority class, we ensure thorough exploration of the majority class decision space with consistent reinforcement of the minority class.

Algorithm 1 shows the pseudo-code of our balanced-bagging algorithm. It constructs K base classifiers, where each classifier h_k is trained on a balanced subset $\mathcal{D}_k$ created through controlled sampling. It employs XGBoost [55] as the base learner due to its demonstrated effectiveness in handling non-linear relationships, feature interactions, and gradient-based

optimization. Although XGBoost is itself an ensemble method, we use it as a base learner for several reasons: (i) each XGBoost model sees a different balanced sample of the data, creating diversity at the data level rather than just the model level; (ii) XGBoost's gradient boosting focuses on sequential error correction, while our bagging approach provides parallel diversity through different data distributions; and (iii) the combination addresses both algorithmic bias (through boosting) and sampling bias (through balanced bagging).

Algorithm 1 Balanced Bagging for Heavy-Tailed Loss Distribution

Require: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, ensemble size K, sampling ratio r
Ensure: Ensemble $\mathcal{H} = \{h_1, h_2, \ldots, h_K\}$

```
Partition D into D_maj and D_min based on class labels
for k = 1 to K do
    Sample ⌊|D_min| · r⌋ instances from D_maj to form D_maj^(k)
    Create balanced subset D_k = D_min ∪ D_maj^(k)
    Train base classifier h_k on D_k using XGBoost with fixed hyperparameters
    Add h_k to ensemble H
end for
return H
```

The optimal sampling ratio r was determined through a two-stage optimization process. First, preliminary random search across $r \in [1.0, 20.0]$ identified promising regions, revealing that performance gains decrease significantly beyond $r = 4$ while computational costs increase linearly. Second, systematic grid search across $r \in \{1.17, 1.47, 2.12, 2.80, 3.12, 3.20, 3.60, 3.85\}$ was performed, with each configuration evaluated on the same validation set to ensure consistent comparison. Let $\hat{y}_i(r)$ denote the predicted class for customer i under sampling ratio r, where $\hat{y}_i(r) = 1$ indicates exclusion (high-risk) and $\hat{y}_i(r) = 0$ indicates acceptance (low-risk). The optimization framework employs constrained profit maximization:

$$r^* = \arg\max_r \mathbb{E}[\text{Profit}_{\text{validation}}(r)] \tag{1}$$

$$\text{subject to} \quad \frac{|\{i : \hat{y}_i(r) = 1\}|}{n_{\text{validation}}} \leq 0.08 \tag{2}$$

The constraint ensures that the proportion of customers flagged for exclusion does not exceed 8% of the validation set. This constraint applies to the total exclusion rate (both correctly identified high-risk customers and incorrectly flagged profitable customers). Confidence intervals were computed using non-parametric (bootstrap) methods. Statistical significance was assessed using Friedman tests for multiple comparisons ($\chi^2 = 98.67$, with $p < 0.0001$) and Wilcoxon signed-rank tests for pairwise comparisons ($\alpha = 0.05$). The profit function calculates realized benefit by omitting customers classified as high-loss (class 1) and summing the values of average annual benefit for selected customers (class 0):

$$\text{Profit}(r) = \sum_{i \in \mathcal{I}_{selected}(r)} \text{avg. annual profit}_i$$

where $\mathcal{I}_{selected}(r) = \{i : \hat{y}_i(r) = 0\}$ represents the set of customers selected under sampling ratio r. This profit-centric objective directly aligns with business value creation, while the constraint ensures adherence to operational limitations imposed by competitive market positioning and regulatory compliance requirements.

Final predictions are generated through majority voting among the K base classifiers:

$$\hat{y} = \arg\max_{c \in \{0,1\}} \sum_{k=1}^{K} \mathbf{1}[h_k(x) = c]$$

where $\mathbf{1}[\cdot]$ denotes the indicator function. This aggregation strategy provides natural uncertainty quantification through vote distribution and enhances robustness against individual classifier errors. The ensemble size $K = 50$ was selected based on convergence analysis showing that additional base learners beyond this threshold provide marginal performance improvements while increasing computational overhead.

4.2. Transformer-Based Model

The proposed deep learning model is based on the Transformer encoder architecture, composed of three layers with an embedding dimension of 128 and a feedforward dimension of 512. The input data, consisting of 194 features, is first projected into a 128-dimensional space through a linear layer. These embeddings are then processed by the Transformer encoder, and the output is passed through another linear layer that performs the final classification. This classification layer also includes a dropout mechanism, intended both to reduce overfitting during training and to improve the model's generalization to unseen data.

Moreover, the Transformer does not output binary labels (0 for good clients and 1 for potential high-loss customers) but rather continuous values in the range $(-\infty, +\infty)$. Clients with scores below zero are considered good, whereas those with scores above zero are classified as potentially bad. To satisfy the business constraint that no more than 8% of clients can be rejected, the model output is treated as a ranking, and the 8% of clients with the highest scores are thus identified as rejected cases. This architecture is simpler than other Transformer-based models for tabular data, such as the FT-Transformer introduced in Gorishniy et al. [46]. However, as explained in previous sections, training speed was a key requirement for the company, motivating the use of a lightweight model. The architecuture used can be seen in Figure 2.

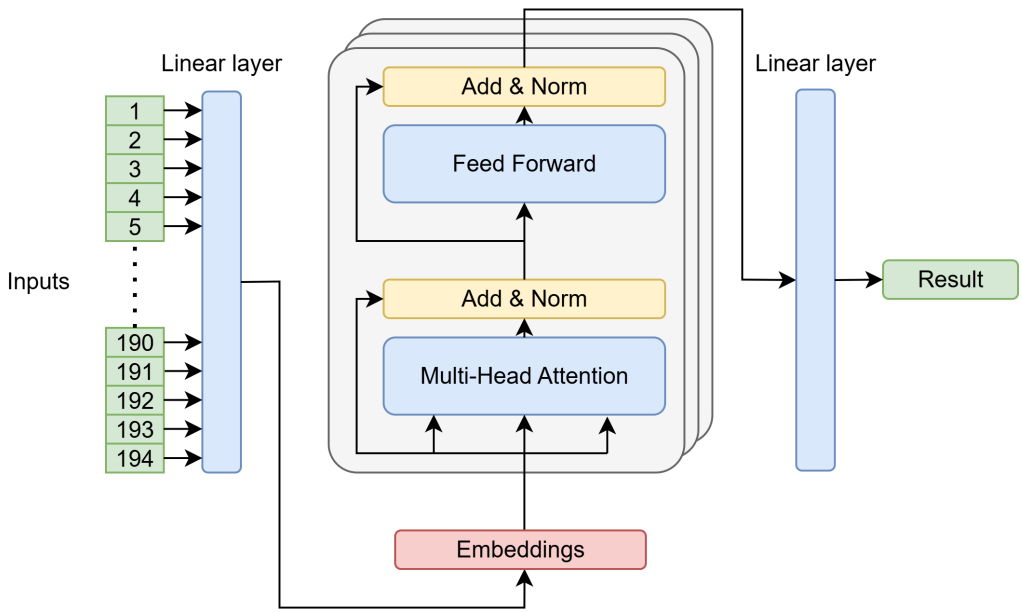

Figure 2. Architecture of the Transformer approach.

For the training data, the class imbalance is addressed by oversampling instances of potentially bad clients. Specifically, the training set is augmented by repeating existing samples of bad clients until a ratio of 3.5 between good and bad clients is achieved. This procedure ensures that the model is exposed to a sufficient number of high-risk examples during training, and the value of 3.5 was determined empirically based on experimentation. The model was then trained for 25 epochs with a learning rate of $5 \cdot 10^{-5}$ using the Adam optimizer (Kingma [56]), and after each epoch, the profit on the validation set was

computed. Among all epochs, the model that achieved the highest validation profit was selected, and it was subsequently used to generate the final results.

4.2.1. Methodoloy Overview

To improve readability, Figure 3 provides a visual summary of the proposed methodology. It shows the main steps of the process, from data preparation and feature selection to the application of business constraints, training of the balanced bagging ensemble, and final decision-making through majority voting. In addition, the diagram incorporates the Transformer-based model as a parallel workflow, illustrating how both approaches share the same data preparation pipeline but diverge into two distinct modeling strategies—one based on balanced bagging and one based on deep representation learning. This diagram complements the pseudo-code (Algorithm 1), the Transformer training description, and the mathematical formulation by providing a unified visual overview of the complete workflow for both models.

Figure 3. Diagram of the proposed methodology, illustrating the shared data preparation pipeline and the two modeling approaches developed in parallel: the balanced bagging ensemble and the Transformer-based model.

5. Results

5.1. Balanced Ensemble Model

5.1.1. Sampling Strategy Optimization Results

The random search optimization evaluated eight different sampling strategies across the range $r \in [1.17, 3.85]$, with each configuration assessed using the same hold-out validation set to ensure consistent comparison. Bootstrap resampling with replacement with $1,000$ iterations provided robust confidence interval estimation for all performance metrics. Table 3 presents a performance analysis for each sampling ratio.

Table 3. Random search results for sampling strategy optimization.

Sampling Strategy	Mean Real Benefit (euros)	Bootstrap 95% CI (euros)	Clients Omitted (%)	F1-Score (mean)	Constraint Compliant
1.17	727,833	(687621, 766880)	27.6	0.797	No
1.47	713,699	(674201, 754649)	20.5	0.836	No
2.12	616,093	(561111, 671588)	12.4	0.874	No
2.80	536,471	(472676, 595574)	8.2	0.890	No
3.12	505,111	(436844, 566107)	7.0	0.894	Yes
3.20	489,752	(422913, 553440)	6.8	0.894	Yes
3.60	460,843	(399643, 526616)	5.7	0.897	Yes
3.85	435,281	(382279, 491054)	5.1	0.898	Yes

The Friedman test confirmed significant variation across strategies ($\chi^2 = 98.67$, $p < 0.0001$). While sampling ratios 1.17 and 1.47 achieved the highest absolute profits ($727,833$ and $713,699$ euros), they violated the constraint by omitting 28% and 21% of customers. Among compliant strategies ($\leq 8\%$ omission), $r^* = 3.12$ achieved the highest benefit ($505,111$ euros, 95% CI: $(436844, 566107)$), with 7.0% omission and F1-score of 0.894. Sampling ratio 2.80, despite higher profit ($536,471$ euros), violated the constraint at 8.2% omission. With 50 models each sampling $7,172$ majority instances from a pool of $26,606$, all majority class instances appeared at least once (average frequency: 13.5 times per instance).

5.1.2. Model Performance on the Test Set

The optimized ensemble with $r = 3.12$ was evaluated on the test set ($15,486$ customers). Figure 4 and Table 4 summarize performance metrics. The model achieves high performance on the majority class (F1: 0.94) but limited minority class recall (30%, F1: 0.32), reflecting the inherent challenge of detecting high-risk customers under severe imbalance.

Table 4. Classification report of the balanced bagging model on the test set.

Class	Precision	Recall	F1-score	Support
False	0.94	0.94	0.94	14250
True	0.32	0.30	0.31	1236
Accuracy			0.89	15486
Macro avg	0.63	0.62	0.63	15486
Weighted avg	0.89	0.89	0.89	15486

The confusion matrix revealed $13,463$ true negatives, 376 true positives, 787 false positives, and 860 false negatives. Performance metrics remained consistent with validation results. The model achieved a real benefit of $1,459,411$ euros (40% of maximum), corresponding to 101.89 euros per customer while omitting 7.51% of customers.

5.1.3. Economic Impact and Misclassification Analysis

The detailed analysis of test set predictions reveals economic trade-offs that are inherent in customer classification under business constraints. Table 5 presents the comprehensive breakdown of misclassification costs and their business implications.

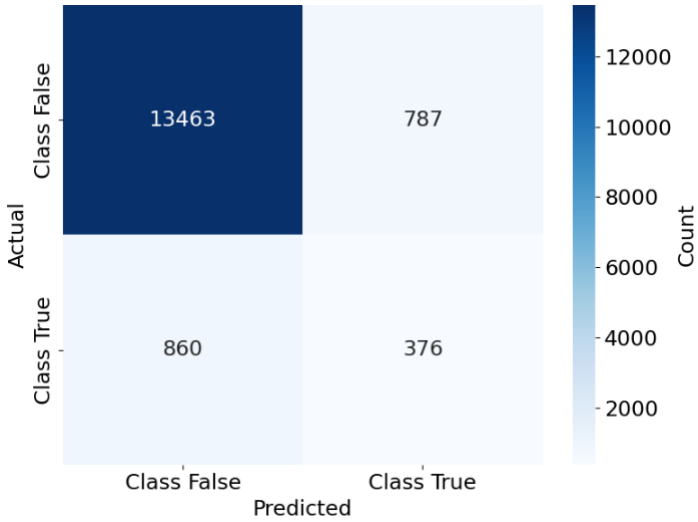

Figure 4. Confusion matrix of the balanced bagging model on the test set, showing 31% recall for high-risk customers.

Table 5. Economic impact analysis by customer category.

Category	Count	Mean Profit (eu)	Total Impact (eu)	Business Implication
True Negatives	13,463	245.40	$+3,303,887$	Correctly Selected
True Positives	376	$-3,007.55$	$-1,130,838$	Correctly Omitted
False Negatives	860	$-2,144.74$	$-1,844,476$	Missed high-risk
False Positives	787	98.39	$+77,434$	Foregone profit

False negatives impose higher per-customer costs ($-2,145$ euros) than false positives (98 euros), validating the recall-prioritized approach. The ten worst false negatives account for 24.4% of total false negative losses ($-481,889$ euros), with the worst case at $-62,398$ euros, demonstrating that extreme cases remain challenging despite overall good performance. False positives show more distributed impact: the ten highest-value omitted customers represent 23.3% of foregone profit (17,534 euros), with maximum individual cost of 4,393 euros. True positives average $-3,008$ euros loss per customer, while true negatives average 245 euros profit, yielding a 3,253 euros differential. False negatives concentrate in the $-1,000$ to $-2,000$ range, indicating systematic challenges in distinguishing moderate-loss from profitable customers.

5.2. Transformer-Based Model

5.2.1. Augmentation Strategy Optimization Results

For the Transformer approach, the augmentation ratio did not affect compliance with the 8% omission constraint, since the model outputs a continuous ranking and always removes the top 8% highest-risk clients. Consequently, the sampling ratio was tuned only with respect to validation profit. A search was performed over ratios between 2.8 and 4.0, using increments of 0.1, and each value was evaluated using the same validation set to ensure a consistent comparison across configurations. For every ratio, the model was trained and the corresponding validation profit was recorded, allowing a direct assessment of the effect of each oversampling level on performance. This procedure indicated that a ratio of 3.5 achieved the highest validation profit, and this value was therefore selected for training the final model. Values outside the range of 2.8 to 4.0 produced poor performance, so the search was concentrated within this interval.

5.2.2. Model Performance on Test Set

The Transformer model trained with an augmentation ratio of $r = 3.5$ was evaluated on the test set (15,486 customers), and Table 6 summarizes the resulting classification performance. The model performs strongly on the majority class (F1-score: 0.94), which, as seen in the previous section, is expected given the dominant proportion of low-risk customers. In contrast, performance on the minority (high-risk) class is substantially lower (Recall: 0.26, F1-score: 0.27), showing the difficulty of identifying rare high-risk cases under severe class imbalance for the Transformer model, with results that are similar but lower than the ensemble method. However, overall accuracy (0.88) and the weighted metrics remain high due to the prevalence of the majority class. Both methods exhibit a similar pattern: high performance on the majority class and limited recall on the minority class.

Table 6. Classification report of the Transformer model on the test set.

Class	Precision	Recall	F1-score	Support
False	0.94	0.94	0.94	14250
True	0.27	0.26	0.27	1236
Accuracy			0.88	15486
Macro avg	0.60	0.60	0.60	15486
Weighted avg	0.88	0.88	0.88	15486

The confusion matrix in Figure 5 shows 13,382 true negatives, 323 true positives, 868 false positives, and 913 false negatives. The performance metrics were consistent with those observed on the validation set and in the ensemble method. Using the model to exclude 8% of customers with the highest predicted risk, a real benefit of €1,378,241 was achieved, corresponding to 38% of the maximum.

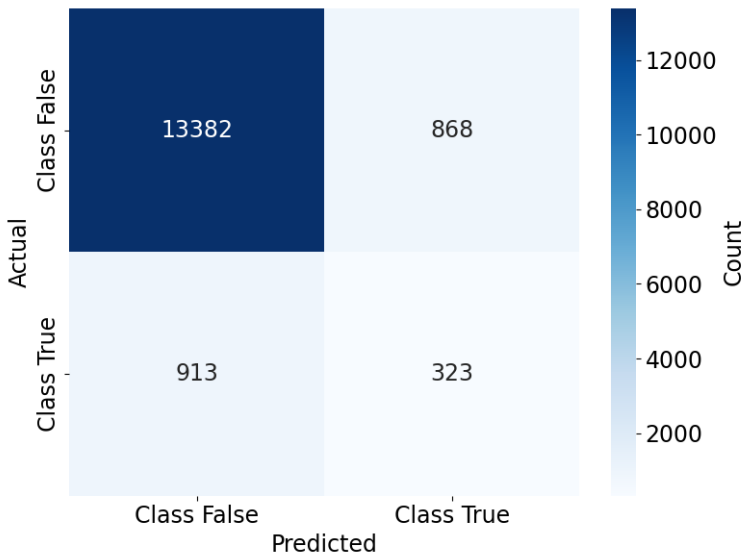

Figure 5. Confusion matrix of the Transformer model on the test set.

5.2.3. Economic Impact and Missclasification Analysis

Similar to the ensemble method, the economic impact analysis by customer category for the Transformer model shows the asymmetric costs associated with classification errors. False negatives, corresponding to high-risk clients that were not detected, result in the largest losses per customer ($-2,061$ euros) and a total impact of $-1,869,073$ euros, confirming that failing to identify these clients represents the greatest financial risk. In comparison, the ensemble method produced slightly higher per-customer costs for false negatives ($-2,145$ euros), but both approaches face challenges with the most costly high-

risk clients. False positives, which are low-risk clients incorrectly omitted, have a much smaller average loss (147 euros) and total impact ($+134,007$ euros). True positives, correctly omitted high-risk clients, total $-1,106,242$ euros, while true negatives generate a positive total impact of $+3,247,314$ euros, closely matching the profit-loss differential seen with the ensemble. Overall, these results indicate that the Transformer model effectively captures profitable clients while showing that a relatively small number of missed high-risk customers can produce significant economic consequences, similar to the ensemble method.

Table 7. Economic impact analysis by customer category.

Category	Count	Mean Profit (eu)	Total Impact (eu)	Business Implication
True Negatives	13,463	243.41	$+3,247,314$	Correctly Selected
True Positives	376	$-3,362.44$	$-1,106,242$	Correctly Omitted
False Negatives	860	$-2,060.72$	$-1,869,073$	Missed high-risk
False Positives	787	147.42	$+134,007$	Foregone profit

6. Discussion

To evaluate the effectiveness of our balanced ensemble approach, the results are compared with the methodology proposed by Soriano-Gonzalez et al. [6], who applied traditional gradient boosting methods (XGBoost and LightGBM) with early stopping to the same customer classification problem.

6.1. Comparative Analysis using a Real Dataset

Table 8 presents a comprehensive comparison between our balanced ensemble approach and the aforementioned baseline methodology.

Table 8. Performance comparison: balanced ensemble vs. baseline methodology.

Metric	Soriano-Gonzalez	Balanced Ensemble	Improvement
ROC-AUC (Test)	0.72	0.90	$+25.0\%$
Precision (High-Risk)	0.67	0.32	-52.2%
Recall (High-Risk)	0.23	0.30	$+30.4\%$
F1-Score (Weighted)	0.79	0.89	$+12.7\%$
Customer Omission Rate	6.0%	7.5%	$+1.5pp$
Business Metrics			
Test Set Benefit	1,232,663 euros	1,459,411 euros	$+18.4\%$
Avg. Benefit per Customer	85 euros	102 euros	$+20.0\%$
Benefit as % of Maximum	33%	40%	$+7pp$

The comparison between our balanced ensemble and the baseline methodology of Soriano-Gonzalez et al. [6] reveals substantial differences in predictive and economic performance. While the baseline achieves a reasonable level of discrimination (ROC-AUC $= 0.72$), the balanced ensemble significantly improves global predictive ability (ROC-AUC $= 0.90$, $+25.0\%$), confirming the advantage of explicitly addressing class imbalance through systematic undersampling and ensemble aggregation. This improvement is also reflected in the weighted F1-score (0.89 vs. 0.79), indicating a more favorable balance between precision and recall at the overall level.

The two approaches differ in their treatment of the minority class. The baseline achieves higher precision for high-risk customers (67% vs. 32%), reflecting better control of false positives. Conversely, the balanced ensemble attains substantially higher recall (30% vs. 23%), identifying a greater proportion of true high-risk customers. Given the asymmetric cost structure of the problem—where false negatives carry much higher financial impact than false positives—the improved recall of the ensemble is more aligned with business objectives. Both approaches respect the operational omission constraint, but the balanced

ensemble produces a significantly higher benefit on the test set (1.46 million euros, +18.4% over the baseline).

From a business perspective, these improvements translate into higher average benefit per customer (102 euros vs. 85 euros) and a greater proportion of the achievable maximum benefit (40% vs. 33%). Although the baseline approach is more precise, the ensemble's superior recall and higher economic return make it a more effective strategy under asymmetric cost conditions typical of insurance applications. Overall, the balanced ensemble delivers a more favorable trade-off between predictive performance and business impact, achieving substantial economic gains while maintaining operational feasibility.

Table 9 presents a detailed comparison between the Transformer-based model and our balanced ensemble approach. While both methods achieve comparable overall classification performance, their behaviors differ substantially across metrics relevant to business and operational objectives.

Table 9. Performance comparison: Soriano–González methodology vs. Transformer model.

Metric	Soriano–González	Transformer	Improvement
ROC-AUC (Test)	0.72	0.71	−1.4%
Precision (High-Risk)	0.67	0.27	−61.1%
Recall (High-Risk)	0.23	0.26	+13.0%
F1-Score (Weighted)	0.79	0.88	+11.3%
Customer Omission Rate	6.0%	8.0%	$+2.0pp$
Business Metrics			
Test Set Benefit	1,232,663 euros	1,378,241 euros	+11.8%
Avg. Benefit per Customer	85 euros	96 euros	+12.9%
Benefit as % of Maximum	33%	38%	$+5pp$

The Transformer model exhibits mixed performance when compared with the baseline methodology of Soriano-Gonzalez et al. [6]. In terms of global discrimination ability, the baseline attains a slightly higher ROC-AUC (0.72 vs. 0.71), indicating a marginal advantage in separating high-risk from low-risk customers. However, the Transformer achieves higher recall for the high-risk segment (26% vs. 23%), identifying a larger proportion of truly risky cases. Precision decreases substantially (27% vs. 67%), reflecting a higher number of false positives, but this trade-off aligns with the model's tendency to prioritize sensitivity over strict error control in the minority class. The weighted F1-score improves from 0.79 to 0.88, suggesting a better overall balance between precision and recall at the global level.

Operationally, the Transformer displays a higher customer omission rate (8.0% vs. 6.0%), though it remains within the predefined 8% constraint. From a business perspective, the Transformer yields significantly higher economic value, increasing total benefit on the test set by 11.8% (1,378,241 euros vs. 1,232,663 euros) and improving the average benefit per customer (96 euros vs. 85 euros). The proportion of the maximum achievable benefit also rises from 33% to 38%, indicating greater economic efficiency despite the higher rate of false positives.

Overall, while the baseline methodology preserves superior precision and slightly better discrimination capability, the Transformer delivers stronger economic performance due to its improved recall and higher profitability. This trade-off reflects the Transformer's ability to capture more high-risk customers at the expense of additional false positives, a pattern that may be advantageous under asymmetric cost structures where missed detections are substantially more costly than false alarms.

6.2. Comparative Analysis with Baseline Methodology using a Synthetic Dataset

To assess the stability of the models under controlled perturbations and evaluate their generalization capacity in reproducible scenarios, nine synthetic versions of the original dataset were generated. Random noise was added to the continuous numerical variables following a normal distribution with zero mean and a standard deviation proportional to the range of each variable. The resulting values were then clipped to remain within their

original limits, preserving the statistical structure and inter-variable relationships while ensuring data coherence and protecting the integrity of the original information. As shown in Table 10, the index appended to each dataset name indicates the level of noise introduced into the original data, where index 0 corresponds to the unaltered dataset and higher indices represent increasing amounts of noise. Each synthetic dataset was used to train and evaluate three modeling approaches: the original methodology proposed by Soriano-Gonzalez et al. [6], based on gradient boosting with early stopping; the Transformer-based model designed for financial risk prediction; and our balanced ensemble approach (bagging boosting), explicitly developed to address class imbalance through systematic undersampling and ensemble aggregation.

For each model and dataset, three indicators were computed: total benefit on the test set, benefit percentage relative to the theoretical maximum, and average training time. The results, summarized in Table 10, reveal substantial differences among the three approaches. The Transformer model achieves high benefits (ranging between 1.1 and 1.4 million euros) with an average percentage of 34%, but at the cost of very high training times (approximately 240–260 s). The balanced ensemble approach obtains comparable results in terms of benefit (around 1.23 million euros and 33% on average) while significantly reducing computation time (approximately 40 to 45 seconds). In contrast, the original methodology yields substantially lower benefits (around 0.8 million euros and 22%) but with nearly instantaneous training times. These findings highlight the trade-off between performance and computational efficiency across models and emphasize the balanced ensemble's advantage in achieving a favorable compromise between accuracy, robustness, and efficiency.

Table 10. Performance comparison across synthetic datasets for the three modeling approaches.

Dataset	Transformer			Soriano-Gonzalez et al. [6]			Balanced Ensemble		
	Benefit (eu)	%	Train Time (s)	Benefit (eu)	%	Train Time (s)	Benefit (eu)	%	Train Time (s)
Synthetic000	1,378,241	37	250.14	1,232,663	33	0.11	1,459,411	40	40.59
Synthetic001	1,418,905	39	235.84	940,327	26	0.11	1,214,087	33	38.09
Synthetic003	1,363,697	37	245.82	1,160,155	32	0.21	1,477,696	40	44.77
Synthetic005	1,206,787	33	237.68	1,111,400	30	0.19	1,250,344	34	46.41
Synthetic008	1,274,579	35	240.92	482,108	13	0.11	1,202,612	33	43.86
Synthetic010	1,268,976	35	239.32	938,804	26	0.21	1,109,951	30	46.51
Synthetic030	1,133,961	31	259.79	639,825	17	0.16	827,443	23	52.99
Synthetic050	1,111,445	30	243.68	406,006	11	0.09	890,799	24	37.06
Synthetic070	1,081,369	29	239.89	508,981	14	0.14	782,906	21	37.48
Synthetic090	1,112,750	30	269.36	482,103	13	0.10	857,873	23	35.19

To statistically assess whether the performance differences observed among the three modeling approaches were significant, we conducted both parametric and non-parametric analyses across the synthetic datasets. A repeated-measures ANOVA revealed a strong effect of the modeling strategy on the achieved benefit ($p < 0.001$), indicating that the three approaches differ substantially. To ensure robustness to distributional assumptions, a complementary Friedman test confirmed the same pattern ($p < 0.001$, Kendall's $W = 0.79$), demonstrating that the observed differences remain consistent even under non-parametric evaluation.

Pairwise comparisons showed a highly consistent structure across both analysis families: the Balanced Ensemble model and the Transformer-based model significantly outperformed the original boosting methodology ($p < 0.01$ in all tests). No statistically significant difference was observed between Balanced Ensemble and Transformer ($p > 0.19$), indicating that both advanced approaches yield comparable benefits. These results confirm that modern architectures provide large quantitative gains over the baseline method. While the Transformer displayed slightly higher mean benefits, the Balanced Ensemble achieved similar performance with markedly lower computational cost, making it more suitable for practical deployment.

Figure 6 illustrates the distribution of benefits across synthetic datasets for the three modeling strategies. The Transformer and the Balanced Ensemble not only achieve higher benefit values than the original approach, but also exhibit noticeably greater stability across increasing noise levels. In particular, the Transformer shows the smallest dispersion,

suggesting greater robustness to perturbations and improved generalization under synthetic data variations. This reinforces the statistical results, highlighting that both modern approaches are substantially more resilient and effective than the original methodology.

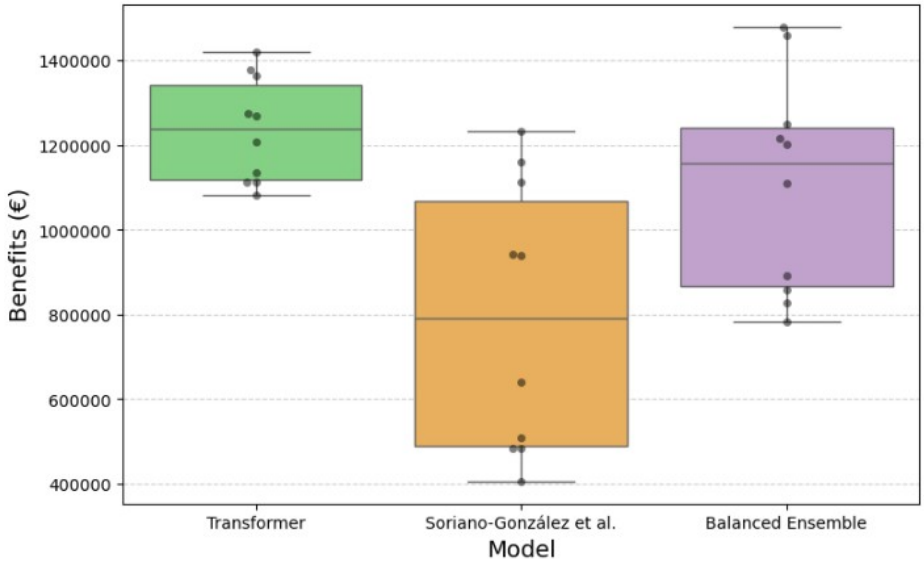

Figure 6. Distribution of benefits across synthetic datasets for the three modeling strategies.

7. Conclusion and Future Work

This study presented a classification framework for new insurance customers without prior historical data, comparing two complementary approaches: a balanced XGBoost-based ensemble and a lightweight Transformer model for tabular data. Both methods operate within a profit-oriented optimization scheme that explicitly enforces the operational constraint of limiting customer omission to 8%.

The empirical analysis shows that the balanced ensemble achieves the best overall trade-off between predictive performance, interpretability, and computational efficiency, yielding higher economic returns and improved recall of high-risk customers. Conversely, the Transformer model delivers competitive results and exhibits superior robustness under perturbed or synthetic variants of the data, highlighting its potential for generalization in more complex or evolving environments. These results indicate that the two approaches are complementary: the ensemble provides a reliable and easily deployable solution in regulated settings, while the Transformer offers a flexible alternative whose performance is expected to scale with the availability of additional data and computational resources.

Overall, the findings demonstrate that it is possible to develop classification models that effectively balance predictive accuracy with business-driven objectives, mitigating loss concentration without compromising portfolio growth. The explicit incorporation of operational constraints and asymmetric cost structures represents a meaningful step towards the practical adoption of machine learning for risk-based decision-making in the insurance sector.

Future work will focus on reducing computational demands, exploring cost-sensitive adaptations to improve precision, and extending validation to diverse markets and regulatory environments. Beyond auto insurance, the methodology is directly transferable to life, health, and reinsurance portfolios, as well as related industries with imbalance challenges such as fraud detection, credit risk, and rare-event prediction in medicine. This broad applicability highlights the potential impact of the proposed approach across domains.

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Abbreviations

The following abbreviations are used in this manuscript:

AUC	Area Under the Curve
CI	Confidence Interval
ML	Machine Learning
MSE	Mean Squared Error
ROC AUC	Receiver Operating Characteristic – Area Under the Curve
SINCO	Information System of the Insurance Compensation Consortium in Spain
SMOTE	Synthetic Minority Over-sampling Technique

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